90-DAY DSA GRIND PLAN

One Problem a Day · ML & Data Science Students

90	90	3
Days	**Problems**	**Phases**
1 per day	Easy → Hard	A · B · C

● PHASE A ■■■■■■■■■■■■	● PHASE B ■■■■■■■■■■■■	● PHASE C
Days 1–30 Foundations	Days 31–60 Core Patterns	Days 61–90 Graphs + DP

30–45 min/day · 1 commit minimum · No zero days

Solve → write a 5-line markdown note → commit both files. Every single day.

PHASE A — Days 1–30

Foundations: Arrays, Hashing, Two Pointers, Sliding Window, Stack, Recursion & Linked List Basics.

Phase	Topics	Problems	Difficulty Focus
A	6	30	Easy → Medium → Hard

ARRAYS & HASHING

Day	Problem	Difficulty	Core Idea
1	Two Sum	Easy	Hash map lookup
2	Contains Duplicate	Easy	Set membership
3	Valid Anagram	Easy	Frequency count
4	Group Anagrams	Medium	Hash by signature
5	Top K Frequent Elements	Medium	Bucket / heap
6	Product of Array Except Self	Medium	Prefix and suffix
7	Longest Consecutive Sequence	Medium	Set expansion

TWO POINTERS

Day	Problem	Difficulty	Core Idea
8	Valid Palindrome	Easy	Left/right scan
9	Two Sum II	Medium	Sorted two pointers
10	3Sum	Medium	Sort + pair search
11	Container With Most Water	Medium	Greedy pointer move

SLIDING WINDOW

Day	Problem	Difficulty	Core Idea
12	Best Time to Buy and Sell Stock	Easy	Track min value
13	Longest Substring Without Repeating Characters	Medium	Window with set/map
14	Longest Repeating Character Replacement	Medium	Frequency window
15	Permutation in String	Medium	Fixed-size window
16	Minimum Window Substring	Hard	Expand / shrink window

STACK

Day	Problem	Difficulty	Core Idea
17	Valid Parentheses	Easy	Stack matching
18	Min Stack	Medium	Auxiliary stack
19	Evaluate Reverse Polish Notation	Medium	Operator stack

20	Generate Parentheses	Medium	Backtracking with constraints
21	Daily Temperatures	Medium	Monotonic stack
22	Car Fleet	Medium	Sort + stack logic
23	Largest Rectangle in Histogram	Hard	Monotonic stack

■ RECURSION & BACKTRACKING

Day	Problem	Difficulty	Core Idea
24	Subsets	Medium	Include/exclude recursion
25	Combination Sum	Medium	Backtracking paths
26	Permutations	Medium	Path building
27	Subset Sum Pattern	Medium	Recursive state thinking

■ LINKED LIST BASICS

Day	Problem	Difficulty	Core Idea
28	Reverse Linked List	Easy	Pointer reversal
29	Merge Two Sorted Lists	Easy	Dummy head merge
30	Linked List Cycle	Easy	Fast and slow pointers

✓ Phase Complete 6 topics · 30 problems *Checkpoint → re-solve 2 Medium without notes*

PHASE B — Days 31–60

Core Patterns: Binary Search, Linked List, Trees, Heap / Priority Queue & Graph Basics.

Phase	Topics	Problems	Difficulty Focus
B	5	30	Medium → Hard

■ BINARY SEARCH

Day	Problem	Difficulty	Core Idea
31	Binary Search	Easy	Classic lo/hi
32	Search a 2D Matrix	Medium	Index mapping
33	Koko Eating Bananas	Medium	Binary search on answer
34	Find Minimum in Rotated Sorted Array	Medium	Pivot reasoning
35	Search in Rotated Sorted Array	Medium	Sorted half check
36	Time Based Key-Value Store	Medium	Binary search on timestamps

■ LINKED LIST

Day	Problem	Difficulty	Core Idea
37	Reorder List	Medium	Split + reverse + merge
38	Remove Nth Node From End of List	Medium	Two pointers
39	Copy List with Random Pointer	Medium	Hash map links
40	Add Two Numbers	Medium	Carry propagation
41	LRU Cache	Hard	Hash map + doubly linked list

■ TREES

Day	Problem	Difficulty	Core Idea
42	Invert Binary Tree	Easy	Recursive swap
43	Maximum Depth of Binary Tree	Easy	DFS depth
44	Diameter of Binary Tree	Easy	Height-based DFS
45	Balanced Binary Tree	Easy	Height validation
46	Same Tree	Easy	Recursive compare
47	Subtree of Another Tree	Easy	Tree matching
48	Lowest Common Ancestor of BST	Medium	BST ordering
49	Binary Tree Level Order Traversal	Medium	BFS queue
50	Binary Tree Right Side View	Medium	Level traversal
51	Count Good Nodes in Binary Tree	Medium	Path maximum
52	Validate Binary Search Tree	Medium	Range constraints

53	Kth Smallest Element in a BST	Medium	Inorder traversal
54	Construct Binary Tree from Preorder and Inorder	Medium	Recursive partition

■ HEAP / PRIORITY QUEUE

Day	Problem	Difficulty	Core Idea
55	Kth Largest Element in an Array	Medium	Min heap
56	Top K Frequent Words or Elements	Medium	Heap ordering
57	Find Median from Data Stream	Hard	Two heaps

■ GRAPHS BASICS

Day	Problem	Difficulty	Core Idea
58	Number of Islands	Medium	Grid DFS/BFS
59	Clone Graph	Medium	Visited map
60	Max Area of Island	Medium	Flood fill count

✓ Phase Complete 5 topics · 30 problems *Checkpoint → re-solve 2 Medium without notes*

PHASE C — Days 61–90

Graphs + DP Essentials: Advanced graphs, 1D DP, 2D DP, Strings & Matrices, and final stretch.

Phase	Topics	Problems	Difficulty Focus
C	5	30	Medium → Hard

■ GRAPHS

Day	Problem	Difficulty	Core Idea
61	Pacific Atlantic Water Flow	Medium	Reverse reachability
62	Surrounded Regions	Medium	Border DFS
63	Rotting Oranges	Medium	Multi-source BFS
64	Walls and Gates	Medium	Distance BFS
65	Course Schedule	Medium	Cycle detection / topo sort
66	Course Schedule II	Medium	Topological ordering
67	Redundant Connection	Medium	Union-Find
68	Graph Valid Tree	Medium	Connectivity + cycle check
69	Word Ladder	Hard	Shortest transformation BFS

■ DYNAMIC PROGRAMMING 1D

Day	Problem	Difficulty	Core Idea
70	Climbing Stairs	Easy	State transition
71	Min Cost Climbing Stairs	Easy	Small DP table
72	House Robber	Medium	Take vs skip
73	House Robber II	Medium	Two linear runs
74	Longest Palindromic Substring	Medium	Expand around center
75	Palindromic Substrings	Medium	Count palindromes
76	Decode Ways	Medium	Index DP
77	Coin Change	Medium	Minimum states
78	Maximum Product Subarray	Medium	Track max/min

■ DYNAMIC PROGRAMMING 2D

Day	Problem	Difficulty	Core Idea
79	Unique Paths	Medium	Grid DP
80	Longest Common Subsequence	Medium	Classic 2D DP
81	Best Time to Buy & Sell Stock with Cooldown	Medium	State machine DP
82	Coin Change II	Medium	Combinations count

83	Target Sum	Medium	Subset transformation
84	Interleaving String	Medium	2D state check

■ STRINGS & MATRICES REVIEW

Day	Problem	Difficulty	Core Idea
85	Longest Common Prefix	Easy	Horizontal scan
86	Valid Sudoku	Medium	Row/col/box hashing
87	Set Matrix Zeroes	Medium	In-place markers
88	Spiral Matrix	Medium	Boundary traversal

■ MATH & BIT — FINAL STRETCH

Day	Problem	Difficulty	Core Idea
89	Pow(x, n)	Medium	Fast exponentiation
90	Single Number	Easy	XOR trick

✓ Phase Complete 5 topics · 30 problems *Checkpoint → re-solve 2 Medium without notes*